

# The necessity of identity

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PHIL UA-78 Metaphysics

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## Leibniz's Law

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## The logic of numerical identity (again)

**Reflexivity** Everything is identical to itself.

**Leibniz's Law** If  $x$  is identical to  $y$ ,  $y$  has every property that  $x$  has.

## A note on Leibniz's Law and tense

NB: Leibniz's Law does NOT say 'If  $x$  is identical to  $y$ ,  $y$  has every property that  $x$  ONCE HAD'.

(Which is good, since that principle seems obviously false. I am identical to myself, but I do not have every property I once had!)

Leibniz's Law is thus compatible with the claim that things can change their properties.

## A tricky argument

1. Clark Kent is identical to Superman (premise).
2. Lois wants to be near Superman (premise).
3. Superman has the property of being someone  
Lois wants to be near (from 2).
4. Clark Kent has every property that Superman has  
(from 1 and LL).
5. Clark has the property of being someone Lois  
wants to be near (from 3 and 4).
6. Lois wants to be near Clark (from 5).

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4. Clark Kent has every property that Superman has (from 1 and LL).
5. Clark has the property of being someone Lois wants to be near (from 3 and 4).
6. Lois wants to be near Clark (from 5).

Assuming an appropriate story, 1 and 2 seem true and 6 seems false. Does that show we have to give up LL?



Jerry Siegel and Joe Shuster, 'Man or Superman?'

© DC Comics.

## One possible diagnosis

Verbs like 'want' (and 'believe', 'know', 'say', etc.) are special. The truth or falsity of a sentence of the form 'A wants to be  $\phi$ ' depends not just on what property we use ' $\phi$ ' to refer to (and on what person we use 'A' to refer to), but on what words we use to refer to that property.

Very imperfect analogy: 'Lois addressed Superman as such'.

What does that mean in terms of diagnosing the mistake in the argument?

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What does that mean in terms of diagnosing the mistake in the argument?

There are different schools of thought on this, but one natural idea is that the problem is with the transition from 2 to 3 and or from 5 to 6.

Normally we can freely rephrase ' $N$  has the property of being an  $x$  such that  $\dots x \dots$ ' with ' $\dots N \dots$ '. But when this involves moving  $N$  into a special environment that's sensitive to word choice (as well as reference), such rephrasings are illegitimate.

## The necessity of identity

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# The necessity of identity

Two important principles:

**Necessity of Self-identity (NSI)** For any object  $x$ , it is necessary that  $x$  is identical to  $x$ .

**Necessity of Identity (NI)** For any objects  $x$  and  $y$ , if  $x$  is identical to  $y$ , it is necessary that  $x$  is identical to  $y$ .

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There is a really important proof of NI, assuming NSI.

Suppose  $x$  is identical to  $y$ . By NSI, it is necessary that  $x$  is identical to  $x$ ; or in other words,  $x$  has the property of being something such that it is necessary that  $x$  is identical to it. But by Leibniz's Law,  $y$  has every property that  $x$  has; so  $y$  also has the property of being something such that it is necessary that  $x$  is identical to it. In other words, it is necessary that  $x$  is identical to  $y$ .

This applies to any interpretation of 'necessary', including metaphysical necessity.

# The necessity of identity

Although NSI is widely accepted, there are some philosophers who think it's false on the grounds that (e.g.) if your parents had never met, you would not even have been self-identical. But there is a weaker claim whose proof does not rely on NSI:

**Conditional Necessity of Identity (CNI)** For any objects  $x$  and  $y$ , if  $x$  is identical to  $y$ , it is necessary that if  $x$  is identical to  $x$ , then  $x$  is identical to  $y$ .

We can prove this in exactly the same way as NI, except that instead of relying on NSI, we now merely rely on the (entirely uncontroversial) claim that for any object  $x$ , it is necessary that if  $x$  is identical to  $x$ , then  $x$  is identical to  $x$ .

## A bad objection to the Necessity of Identity

The inventor of bifocals is identical to the first Postmaster-General. But it is not necessary that the inventor of bifocals is identical to the first Postmaster-General. So there exist objects  $x$  and  $y$  such that  $x$  is identical to  $y$  but it is not necessary that  $x$  is identical to  $y$ .

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Response: Remember the earlier warning about determiner phrases!

## A bad objection to the Necessity of Identity

Often, when determiner phrases occur in the same sentence as “operator” expressions (like ‘not’ and modals), we get **scope ambiguities**.

- ▶ ‘Everyone hasn’t arrived’ can mean ‘it is not the case that everyone has arrived’ (not > every) or ‘everyone is such that it is not the case that they have arrived’ (every > not). [ $X > Y$  here means ‘ $X$  takes scope over  $Y$ ’]
- ▶ ‘Everyone could have got a prize’ can mean either ‘It could have happened that everyone got a prize’ (could > every) or ‘Everyone is such that they could have got a prize’ (every > could)
- ▶ ‘The inventor of bifocals could have been born in Germany’ can mean either ‘It could have been that someone born in Germany invented bifocals before anyone else’ (could > the) or ‘The inventor of bifocals is someone who could have been born in Germany’ (the > could).

## A bad objection to the Necessity of Identity

If we give both descriptions scope over the modal, 'It is necessary that the inventor of bifocals is identical to the first Postmaster-General' is equivalent to 'The inventor of bifocals and the first Postmaster-General are an  $x$  and  $y$  such that it is necessary that  $x$  is identical to  $y$ '. On this reading it is plausibly true!

And this is the only reading on which it follows from NI.

In general, you can reason from 'For every object  $x$ , ... $x$ ...' and 'there is a unique  $F$ ' to '...the  $F$  ...', but only if you make sure to understand 'the' in the conclusion as taking widest scope.

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- N.B. this response has nothing to do with our earlier response to the 'Lois wants to be near Superman' argument.

## Proper names

According to an old theory that Kripke is widely thought to have demolished, proper names like 'Kamala Harris' are short for definite descriptions—maybe 'the Democratic candidate for president' or 'the famous person called "Kamala Harris"'.

But proper names do not seem to give rise to scope ambiguities when they combine with modals.

- ▶ 'Kamala Harris could have not been Kamala Harris' seems to be unambiguously false.
- ▶ 'The Democratic candidate could have not been the Democratic candidate' has a salient true reading where the first 'the' scopes over 'could' and the second one scopes under it.

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Kripke puts this by saying that proper names are rigid designators. This means that we don't have to worry about the inference from the NI to, e.g., 'If Superman is Clark Kent, then it is necessary that Superman is Clark Kent'.

## Why the necessity of identity matters

We can use the necessity of identity to argue from identity claims to claims of metaphysical necessity.

For example: we can argue from the premise that Superman is Clark Kent to the conclusion that it is metaphysically necessary that everyone who has met Superman has met Clark Kent.

(The argument relies of course on the uncontroversial premise that 'It is metaphysically necessary that everyone who has met Superman has met Superman'.)

## Why the necessity of identity matters

This form of argument becomes much more powerful and interesting when we turn from identity of objects to identity of properties.

For example: we can argue from the premise that the property of being golden is the property of consisting of atoms with 79 protons to the conclusion that it's metaphysically necessary that every gold thing consists of atoms with 79 protons.

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For example: we can argue from the premise that the property of being golden is the property of consisting of atoms with 79 protons to the conclusion that it's metaphysically necessary that every gold thing consists of atoms with 79 protons. And the premise seems plausible! When chemists analysed a bunch of golden things and found that they all consisted of atoms with 79 protons, they got strong evidence for a claim about what it is to be golden—namely, that to be golden is to consist of atoms with 79 protons.

(In my view, Kripke's most important achievement is to make the world safe for this sort of argument, by clearing away various bad objections to it.)